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Question 1 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 5]

The maximum temperature T , in degrees Celsius, on six randomly selected days is shown in the following table. The table also shows the number of soda cans purchased, N , from a vending machine on those six days.

T ($^{\circ}\text{C}$)	5	6	18	32	29	12
N (cans)	26	28	37	41	48	29

1.

The relationship between the variables can be modelled by the regression equation $N = aT + b$.

1.

Find the value of a and the value of b .

2.

Estimate the number of soda cans purchased on a day when the maximum temperature is 23°C .

[4]

2.

Find the value of r , the correlation coefficient.

[1]

Question 2 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 6]

A coffee shop owner believes that the number of coffees they sell on a given day is related to the average temperature of that day. The average temperature T , in degrees Celsius, on seven randomly selected days is shown in the following table. The table also shows the number of coffees, N , the coffee shop sells on those seven days.

T ($^{\circ}\text{C}$)	14.5	16.2	15.3	13.1	10.8	14.7	17.3
N (coffees)	82	54	61	96	112	84	55

1.
The relationship between the variables can be modelled by the regression equation $N = aT + b$.
 1.
Find the value of a and the value of b .
 2.
To the nearest integer, estimate the number of coffees sold on a day when the average temperature is 16°C . [4]
2.
Find the value of r , the correlation coefficient. [2]

Question 3 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 6]

The following table shows the amount of fuel (y litres) used by a car to travel certain distances (x km).

Distance travelled (x km)	14	35	80	115	145	170
Fuel used (y litres)	2.5	6.2	10.8	13.6	15.4	19.7

1.
Find the correlation coefficient. [2]
- The data can be modelled by the regression line with equation $y = ax + b$.
1.
Write down the value of a and the value of b . [2]
 2.
Use the model to estimate the amount of fuel the car would use if it has driven 55 km. [2]

Question 4 Calculator Allowed

Markscheme

 Difficulty: **Easy**

[Maximum mark: 7]

The following table shows the average weights (y kg) for given heights (x cm) in a population of high school males.

Heights (x cm)	160	165	170	175	180	185
Weights (y kg)	62.0	65.9	68.4	71.8	74.6	76.3

1.

The relationship between the variables can be modelled by the regression equation $y = ax + b$.

1.

Write down the value of a and the value of b .

2.

Hence estimate the weight of a male whose height is 167 cm. [4]

2.

1.

Write down the correlation coefficient.

2.

State which **two** of the following describe the correlation between the variables.

negative zero no correlation

strong weak positive

[3]

Question 5 Calculator Allowed

Markscheme

 Difficulty: **Easy**

[Maximum mark: 6]

The following table shows the number of skateboards y that are sold x days after the shop opened.

Time after opening (x days)	2	4	6	8	10	12
Skateboards sold (y)	10	19	28	36	45	55

This data can be modelled by the regression line with equation $y = ax + b$.

1.

1.

Write down the value of a and the value of b .

2.

Explain what the gradient a represents. [4]

2.

Use the model to estimate the number of skateboards sold in 5 days after the shop opened. Round your answer to the nearest integer. [2]

Question 6 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 6]

Extended power failure occurred in John's neighbourhood last winter. He measured the temperature of his house as air inside cooled.

The following table shows air temperature T , in degrees Celsius, t hours after power failed.

t (hours)	0	4	8	12	16	20
T ($^{\circ}\text{C}$)	22.0	18.2	12.5	9.84	4.48	1.15

1.

1.

Write down the independent variable.

2.

Write down the warmest air temperature of the house.

[2]

John believes that the relationship between T and t can be modelled by a linear regression equation.

1.

John describes the correlation as **very strong**. Circle the value below which best represents the correlation coefficient.

[2]

0.997	0.272	0	- 0.272	- 0.997
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2.

John's model is $T = -1.06t + 22.0$, for $0 \leq t \leq 20$. Use this model to predict the decrease in temperature for any 2 hour interval.

[2]

Question 7 Calculator Not Allowed

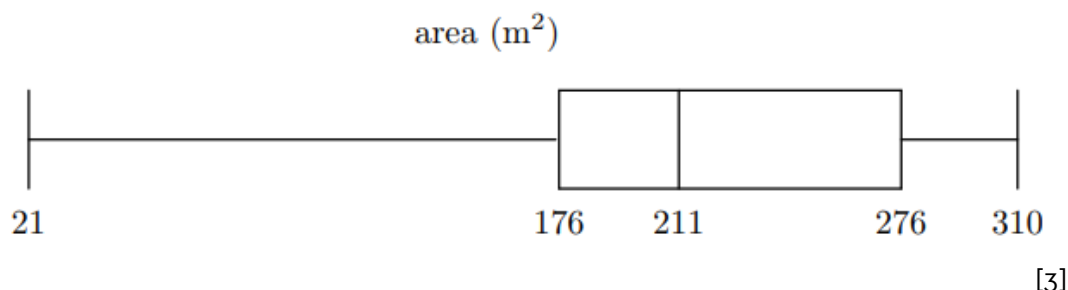
Markscheme

Difficulty: **Easy**

[Maximum mark: 7]

A real estate company keeps a register of the monthly cost of rent, R , of their apartments and their corresponding area, A , in m^2 .

The areas of the apartments registered are summarised in the following box and whisker diagram.



1. Find the smallest area A that would not be considered an outlier.

The regression line A on R is $A = \frac{5}{12}R - 50$.

Meanwhile, the regression line R on A is $R = \frac{5}{2}A + 100$.

- 1.

One of the apartments has a monthly rent of \$480. Estimate the area of the rental.

[2]

- 2.

Find the mean rental cost of all the real estate company's apartments.

[2]

Question 8 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 12]

The following table shows the total revenue, y , in Australian dollars (AUD), obtained daily during the first week of January 2020, by Peppy's Pizza restaurant and the number of guests, x , served.

Number of Guests (x)	45	54	39	51	89	83	90
Revenue in AUD (y)	410	516	423	558	906	843	940

1.
 1. Calculate the Pearson's product-moment correlation coefficient, r , for this data.
 2. Comment on the result. [3]
2. Write down the equation of the regression line y on x . [1]
3. Use the line of the regression to estimate the revenue of serving 70 guests. Give your answer correct to **the nearest AUD**. [2]

The daily maintenance cost for the restaurant is 300 AUD. Additionally, the cost of serving one guest is 5 AUD.

1. Determine if the restaurant expects to make a profit when serving 50 guests on a **particular** day. [2]
2.
 1. Find an expression for the profit of the restaurant when serving x guests on a particular day.
 2. Find the least number of guests required to be served to result in a profit for the day. [4]

Question 9 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 13]

Each day, a factory records the number (x) of memory foam pillows it produces and the total production cost (y) dollars. The results for ten days are shown in the following table.

x	75	88	105	81	124	108	94	112	118	99
y	1805	2022	2264	1834	2690	2338	2144	2485	2542	2256

1.

Write down the equation of the regression line in the form $y = ax + b$ [2]

Use your regression line as a model to answer the following.

1.

Interpret the meaning of:

1.

the gradient;

2.

the y -intercept. [2]

2.

Estimate the cost of producing 100 pillows. [2]

The factory sells the pillows for \$ 29.99 each.

1.

Find the least number of pillows that the factory should produce in one day in order to make a profit. [3]

2.

Comment on the appropriateness of using your model to estimate:

1.

the cost of producing 1000 pillows;

2.

the number of pillows produced when the total production cost is \$1940. [4]

Question 10 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 6]

The following table below shows the points scored by eight basketball players in two games.

Game 1 (x)	5	6	9	12	12	16	20	25
Game 2 (y)	6	8	12	13	15	17	24	22

Let L_1 be the regression line of x on y . The equation of the line L_1 can be written in the form $x = ay + b$.

1.

Find the value of a and the value of b . [2]

Let L_2 be the regression line of y on x . The lines L_1 and L_2 both pass through the same point with coordinates (p, q) .

1.

Find the value of p and the value of q . [2]

2.

Andrew's injury didn't allow him to play in the first game but he played and scored 10 points in the second game. Use an appropriate regression equation to estimate the number of points that Andrew would have scored in the first game. Round your answer to the nearest point. [2]

Question 11 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 6]

The following table below shows the marks scored by six job candidates on two aptitude tests.

Aptitude Test 1 (x)	58	85	96	118	140	132
Aptitude Test 2 (y)	45	64	90	112	120	144

Let L_1 be the regression line of x on y . The equation of the line L_1 can be written in the form $x = ay + b$.

1.

Find the value of a and the value of b . [2]

Let L_2 be the regression line of y on x . The lines L_1 and L_2 both pass through the same point with coordinates (p, q) .

1.

Find the value of p and the value of q . [2]

2.

Chelsea missed the first aptitude test but scored 101 marks on the second aptitude test. Use an appropriate regression equation to estimate Chelsea's mark on the first aptitude test. [2]

Question 12 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 12]

The following table shows values of $\ln x$ and $\ln y$.

$\ln x$	1.24	2.61	4.93	6.55
$\ln y$	4.02	3.29	1.95	1.04

The relationship between $\ln x$ and $\ln y$ can be modelled by the regression equation $\ln y = a \ln x + b$.

1.

Find the value of a and the value of b . [3]

2.

Use the regression equation to estimate the value of y when $x = 4.12$. [3]

The relationship between x and y can be modelled using the formula $y = px^q$, where $p \neq 0$ and $q \neq 0$, $q \neq 1$.

[6]

1. Expressing $\ln y$ in terms of $\ln x$, find the value of p and the value of q .

Question 13 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 15]

A study records the number of chickens and ducks on a farm after t years, starting on 1 January, 1997.

Let c be the number of chickens on the farm after t years. The following table shows the number of chickens after t years.

Number of years (t)	0	3	9	15	20
Number of chickens (c)	112	174	256	330	423

The relationship between the variables can be modelled by the regression equation $c = at + b$, where a and b are constants.

1.
Find the value of a and the value of b . [3]
2.
Use the regression equation to estimate the number of chickens on the farm when $t = 6$. [2]

Let d be the number of ducks on the farm after t years. The number of ducks can be modelled by the equation $d = 400e^{-kt}$, where k is a constant.

1.
Find the number of ducks on the farm on 1 January, 1997. [3]
2.
After six years, there are 377 ducks on the farm. Find the value of k . [3]
3.
Find the year during which the number of chickens and ducks were the same. [4]